

Published in final edited form as:

Curr Biol. 2004 January 6; 14(1): R35–R45. doi:10.1016/j.cub.2003.12.022.

Embryonic Cleavage Cycles: How Is a Mouse Like a Fly?

Patrick H. O'Farrell¹, Jason Stumpff², and Tin Tin Su²

¹ Department Biochemistry and Biophysics, GH-S372C Genentech Hall, UCSF, San Francisco, CA 94143-2200, USA. E-mail: ofarrell@cgl.uscf.edu

² Department Molecular, Cellular and Developmental Biology, 347 UCB, University of Colorado, Boulder, CO 80309-0347

Abstract

The evolutionary advent of uterine support of embryonic growth in mammals is relatively recent. Nonetheless, striking differences in the earliest steps of embryogenesis make it difficult to draw parallels even with other chordates. We suggest that use of fertilization as a reference point misaligns the earliest stages and masks parallels that are evident when development is aligned at conserved stages surrounding gastrulation. In externally deposited eggs from representatives of all the major phyla, gastrulation is preceded by specialized extremely rapid cleavage cell cycles. Mammals also exhibit remarkably fast cell cycles in close association with gastrulation, but instead of beginning development with these rapid cycles, the mammalian egg first devotes itself to the production of extraembryonic structures. Previous attempts to identify common features of cleavage cycles focused on post-fertilization divisions of the mammalian egg. We propose that comparison to the rapid peri-gastrulation cycles is more appropriate and suggest that these cycles are related by evolutionary descent to the early cleavage stages of embryos such as those of frog and fly. The deferral of events in mammalian embryogenesis might be due to an evolutionary shift in the timing of fertilization.

The demands on frog or *Drosophila* eggs, which are deposited in the environment to fend for themselves, are very different from the demands on a mouse egg, which is held in a protective and nutritive environment. Frog and fly eggs need to produce a feeding animal with the reserves within the egg. This produces a cascade of problems and solutions that appears to have become an integral part in the early developmental programs of freely developing organisms [1]. The first problem is to produce a whole feeding organism from an egg. The solution is to make eggs especially large cells to provide adequate reserves. The second problem is that the single allotment of DNA in an egg does not have the capacity to rapidly change the composition of RNAs in the huge cytoplasm of the egg. The solution is to use maternally encoded gene products and to quickly amplify the number of nuclei to provide a transcriptional output that is adequate for the developmental events to come.

The mammalian egg is faced with the very different task of developing a machinery to take advantage of the nutritive environment in which it is located. Thus, it develops extraembryonic tissues for interaction with the uterus and, in doing so, defers the events of early development. Mammalian eggs lack the massive maternal contributions of freely developing eggs, and development begins at a more leisurely pace based largely on zygotic synthesis of components. There is no obvious reason that the mouse egg would need to have especially rapid cleavages, except that a mammal may well rely on developmental programs that evolved during the more than 250,000,000 years of metazoan evolution that preceded the appearance of mammals.

Rapid cleavage cycles are found in all major metazoan phyla, including chordates. Nonetheless, the mammalian embryo begins development with slow divisions and shows rapid cell cycles only at a later stage. Because they do not immediately follow fertilization, these later rapid cycles in mammals are not ordinarily considered homologous to early cleavage cycles of other

embryos. Here, we suggest that fertilization should not be used to align the developmental program of mammals with that of other organisms. Instead, when the highly conserved events surrounding gastrulation are aligned, the rapid division cycles of the mammalian embryo come into correspondence with the cleavage cycles of other metazoan embryos. We summarize evidence suggesting that the mammalian rapid cycles are homologous to the rapid cleavage cycles of other metazoans.

The alignment of embryonic events that we advocate emphasizes that the post-fertilization events of mammalian development begin with the generation of a trophoblast, the key contributor to the mammalian placenta. This process appears to have no analog in the post-fertilization events of non-placental vertebrates. We suggest that these steps may have had an evolutionary precursor in events that contribute to oogenesis in other species. If one considers the maternal events in the oocyte lineage and the zygotic events that follow fertilization as a continuum, a shift in the timing of fertilization with respect to other events occurring in this lineage could shift processes from maternal to zygotic control, or *vice versa*. We propose that, during the evolution of mammals, fertilization was advanced to an earlier stage such that events that occurred late in the cell lineage of the oocyte in the progenitors of mammals were displaced and modified to become the earliest events of post-fertilization development in mammals.

Unique Features of Embryonic Cleavage Cell Cycles

Because *Xenopus* and *Drosophila* are important model systems for cell cycle control, we possess detailed information that allows identification of common features of the early mitotic cycles in chordates and arthropods.

First, pre-gastrulation cycles are unusually fast. A frog egg undergoes six divisions in 3 hours, averaging about 30 minutes per division cycle [2]. A *Drosophila* egg undergoes 13 embryonic cycles in 2 hours with a progressive lengthening of the cycles from 8.3 to 23 minutes [3]. In contrast, proliferating larval tissues have about an 8 hour cell cycle time [4,5].

Second, embryonic cleavage cycles occur without growth, so that cells become progressively smaller. Indeed, this is inevitable, as the embryos have no outside source of nutrition. The progressive reduction in size of cells stands in contrast to most cell cycles, wherein cells grow prior to division to roughly maintain size constancy during proliferation [1,6,7].

Third, the cleavage cycles appear to lack the gap phases that usually intervene between mitosis and S phase, and between S phase and mitosis. Whereas the very first mitotic cycle following fertilization has a short G2 phase in frogs and perhaps also in *Drosophila* [8], the subsequent phases have no detectable gap phases [9,10]. In other words, the very short interphases consist exclusively of S phase. Replication of the entire $1.8 \cdot 10^8$ base pair (bp) genome of *Drosophila* and the $1.7 \cdot 10^9$ bp genome of *Xenopus* [11] is completed in as little as 3.4 and 15 minutes, respectively. This remarkable feat is achieved by the use of many more origins of DNA replication than are active in longer cell cycles [12–15].

Fourth, the embryonic cell cycles of frogs and flies rely on maternally deposited products and can run in the absence of transcription of the zygotic genome. Thus, cell cycle transitions and the regulation of these transitions are independent of transcription.

Fifth, the early cell cycles of *Xenopus* and *Drosophila* appear to lack certain checkpoint controls that ordinarily coordinate progression through the various cell cycle events. Thus, whereas cells from diverse sources (different species, tissues, or cell lines) arrest cell cycle progress when DNA synthesis is blocked, cells progress to mitosis with catastrophic consequences, when DNA replication is blocked by aphidicolin in *Xenopus* and *Drosophila* embryos [16–19]. As a result of these observations, it was initially concluded that the early cycles lacked the

checkpoint controls required to arrest the cells. Newer observations suggest that some checkpoint mechanisms are in place, but in some species are too weak to enforce an arrest [19,20] (see below). The above-described cycles are followed by gastrulation in fly and frog and slower cell cycle times [3,2]. It has been reasoned that early cycles rapidly generate the cells that become fodder for gastrulation and creation of the body layers. Additionally, the exponential increase in the transcriptional capacity has been suggested to be important for the switch to control by zygotic transcription, which occurs in parallel with the completion of the early rapid cycles.

Fast Embryonic Cycles Exist in All Major Animal Phyla

The frog and the fly are model organisms for early development in part because they develop quickly. One important question is whether the organization of the early cycles is more general. While analyses in other systems are less detailed, key features of the early cleavage cycles – their speed, near synchrony, and the progressive decline in cell size – are obvious in descriptive analyses that were pursued widely at the turn of the last century. In his classical book ‘The Cell in Development and Heredity’ [21], E.B. Wilson attributes a generalization that embryogenesis begins with ‘*a series of rapidly succeeding mitotic divisions, thus splitting up [the egg] into blastomeres or embryonic cells*’ to studies of Kölliker and Remak in the mid 19th century. This early evidence of generality is bolstered by more recent and detailed investigations of specific representatives of the major phyla:

Annelids

During the first seven stages of embryogenesis in the leech, *Helobdella triserialis*, early blastomeres contain short cell cycles that lack G1 phases. Following these divisions, primary blast cells cycle, still without a G1 phase, but with a much longer G2 phase [22].

Echinoderms

In the sea urchin, *Paracentrotus lividus*, the first four division cycles are synchronous in all blastomeres and last approximately 30 minutes each. The mitotic index in these embryos is high during the blastula stage (60% of cells are in mitosis in 6 hour old blastulae) and drops dramatically before hatching (11% in hatching blastulas), suggesting a lengthening of interphase [23]. These events precede gastrulation and the overall pattern is consistent with sea urchins exhibiting fast cycles before gastrulation.

Nematodes

In *C. elegans*, cell division patterns are hard-wired, with cells of each lineage dividing with stereotypical timing that is invariant from embryo to embryo [24]. With the exception of the first embryonic cell division, which occurs at about 40 minutes after fertilization, cell cycles that precede gastrulation last 10–30 minutes. Cell cycles lengthen after the onset of gastrulation, although the extent of the lengthening varies between lineages. For example, the first cell cycle after gastrulation ranges from 30–60 minutes in the C-lineage and from 70–90 minutes in the E-lineage.

Molluscs

In the surf clam *Spisula solidissima*, the four mitotic cycles following fertilization last 25–35 minutes each [25,26].

When added to the *Drosophila* and frog models, these examples represent the major phyla in the evolutionary tree of the metazoa: Mollusca (clam), Arthropoda (fruit fly), Annelida (leeches), Echinodermata (Star fish and sea urchin), Chordata (frog) and Nematoda (*C.*

elegans) [27]. It is notable that frogs are not unusual among the Chordata in having rapid cleavage cycles, which are found broadly among birds, amphibians, fish and ascidians.

The early cycles in chick are notable because of the relatively tight evolutionary connection between birds and mammals: like mammals, birds develop an amniotic sac in early development and clear parallels can be drawn between steps of gastrulation in birds and mammals. The first 22.5 hours of post-fertilization chick development occur in the oviduct, as the albumin and shell are deposited. When the egg is laid, the blastodisc has about 60,000 cells [28] (R. Ivarie, personal communication). This requires at least 15 or 16 cell doublings in 22.5 hours and, thus, an average doubling time of less than 1.5 hours. In chick, as in all the characterized examples, gastrulation is tightly coupled with the last stages of the rapid early cleavage cycles.

Integration of Rapid Cell Cycles with Embryonic Patterning

In *Xenopus* and *Drosophila*, the cleavage cycles end with an abrupt transition that is followed by the onset of gastrulation. The transition at the end of the cleavage stages, referred to as the midblastula transition (MBT) in frog and the maternal to zygotic transition (MZT) in flies, is dramatic in that numerous fundamental features of cell behavior change at this time including onset of high level transcription, initiation of cell movements and introduction of a gap phase into the cell cycle. It is not clear why so many changes occur in concert at the end of the cleavage divisions, but there are suggestions that the consuming investment into cellular replication is incompatible with some aspects of morphogenesis and gene expression. Experimental induction of mitosis during gastrulation disrupts the morphogenetic movements [29–31]. Furthermore, rapid cycles interfere with gene expression and the arrest of rapid early cell cycles can advance zygotic transcription [32,33]. Presumably, the longer interphase associated with the introduction of gap phases provides time for cytoskeletal changes that underlie cell movement during gastrulation and opportunity for the relatively time consuming polymerization of lengthy transcripts [32].

The dramatic nature of the MBT/MZT has promoted a simple view in which the cell cycle exists in either embryonic or adult forms, separated by one major embryonic transition. However, the cell cycle has many faces, and even after the MBT/MZT embryos can exhibit very fast cycles. For example, in *Drosophila* the mesoderm, which invaginates and is internalized in the first cell cycle after the MZT, goes through two subsequent cycles that are about 45 minutes long. The neuroblasts have a similarly rapid cycle [34,35]. In both cases, there are no evident gap phases. Thus, during gastrulation and patterning of the embryo, the shortest cycles are five times longer than the cleavage cycles, but are still more than ten times faster than later mitotic cycles in the larval tissues.

It should also be emphasized that the model organisms that we know best are not fully representative. *Drosophila* exemplifies a very successful late branch of arthropod evolution, the long-germ-band insects. Other Arthropods exhibit a more basal developmental mode that is more easily related to events in other phyla. In short-germ-band insects and crustaceans, only the anterior part of the body plan is patterned at the onset of gastrulation and a proliferative group of posterior blast cells supplies cells that build successively more posterior body regions [36]. Thus, whereas a dramatic transition in cell cycle marks the end of the cleavage cycles, local rapid proliferation remains a feature of embryos at gastrulation and later.

Are Mammals Exceptions?

The early cycles following fertilization of the mammalian egg are not unusually fast and appear to resemble more canonical cell cycles. In the mouse, the first cell cycle is long – the fertilized egg reaches the 2-cell stage at 1.5 days post-coitum (dpc). The next four cell cycles average

about 12 hours each leading to the 32 cell early blastocyst at 3.5 dpc [37]. Cell cycles from the early blastocyst stage to implantation of the late blastocyst (~120 cells) take on average about 24 hours. The duration of these cycles is not only comparable to that of typical proliferating cell populations, the cycles also include features lacking in early cleavage cycles. The early mouse cycles have a G1 and a G2, they arrest in response to aphidicolin inhibition of DNA replication and they exhibit a radiation-induced arrest in G2 ([38], see below). As the post-fertilization cycles have little in common with the early cleavage cycles of *Drosophila* or frogs, it is a widely held view that the fast cycles of model organisms are not relevant to mammalian embryonic cell cycle regulation. We agree that the post-fertilization cycles differ, but nonetheless suggest that there are mammalian cell cycles that are homologous to the rapid cleavage cycles of the model organisms, only that these cell cycles are at a different stage.

Aligning Development

There is a discontinuity in the manner in which development of mammals is aligned with that of other organisms. While the earliest stages are aligned based on the use of fertilization as a reference point for the beginning of development, other common features of embryonic patterning have led to an independent alignment of embryogenesis at later stages. The latter is based on remarkably conserved features of morphology, patterning events and expression of conserved genes.

As noted by von Baer and emphasized by Haeckel more than a century ago [21], all vertebrate embryos look remarkably similar after the establishment of body axes, neurulation and the beginning of somite formation (Figure 1A). Similarly, the analysis of embryos of diverse arthropods has identified remarkable similarities after establishment of the body axis, production of a ventral nerve cord and initiation of segmentation [39,40]. This stage has been referred to as the phylotypic stage, because all of the organisms within a phylum appear to resemble each other at this stage. However, at this stage the similarities are not only apparent within a phylum, but also between phyla. Thus, the body plans of arthropods and vertebrates share features such as a central nerve cord, relative position of yolk, gut and nerve cord, and subdivisions along the anterior posterior axis (somites and segments; Figure 1). These morphological parallels are reinforced by parallels in patterns of expression of important determinants of developmental fate (see below). Perhaps we should expect such similarities across phyla at this stage. Just as conserved domains can be recognized in distantly related protein sequences, it is the conserved steps of development that can be most easily recognized in the embryos of distantly related organisms. We argue that, just as the alignment of distantly related proteins is based on conserved domains rather than simply aligning the sequences starting at the amino terminus, so the alignment of distantly related developmental programs ought to be based on alignment of the most conserved stages.

Whereas embryo morphology and size are remarkably conserved at the phylotypic stage, it is commonly recognized that morphology diverges at later stages, as species specific anatomy develops [41]. Furthermore, as the phylotypic stage is the most conserved stage, it should not be a surprise that earlier embryos also show a more highly diverged morphology (Figure 1). Indeed, as one examines progressively earlier stages, homologies between species become gradually less evident. For example, the similarity of neural tube formation via a neural fold is clearly evident in diverse chordates, but central nerve cord formation in other phyla occurs in different ways. Even within chordates, however, the parallels in development become less evident at even earlier stages. Challenging mental gymnastics are required to draw parallels between the gastrulation movements of fish, frog, chick and mouse and the parallels at pregastrulation stages are not clear. Nonetheless, molecular studies and patterns of gene expression have shown that extensive parallels do exist.

Molecular analyses have detected parallels in the gastrulation processes of organisms belonging to different phyla. In organisms from *Drosophila* to human, a conserved set of genes encodes an extracellular signaling pathway that governs dorsal/ventral patterning of the embryo. In this pathway, a BMP type of signaling molecule acts as a ventralizing signal in vertebrates, and, due to a switch in spatial reference-points and hence names, as a dorsalizing signal in arthropods [42,43]. Additionally, other interacting components of this signaling system (e.g., Chordin/Sog, and Twisted gastrulation) are also involved in diverse species [44,45]. Although the pathway remains to be fully elucidated, studies in *Xenopus* have shown that BMP signaling regulates localized expression of the homeodomain protein Goosecoid and the conserved DNA binding protein Brachyury [46,47]. Brachyury and its homologs appear to specify posterior embryonic structures in species extending from *C. elegans* to mammals, whereas Goosecoid and its homologs specify anterior structures [48–52].

While the details of the conservation and mechanism of action of the pattern forming genes are of tremendous importance, we wish to emphasize here their utility in aligning analogous stages of the development of different groups. The BMP signaling cascade in *Xenopus* acts very early, as the egg is undergoing the rapid cleavage cell cycles. The onset of localized expression of Goosecoid and Brachyury precedes gastrulation slightly and persists as a distinctive mark during gastrulation [53]. In species from echinoderms to mice, the expression of these genes shows a similar association with gastrulation [53–57]. Notably, these genes are not expressed in mammalian blastocysts, which are often presented as the analog of the frog blastula (Figure 1,2). Instead, they are expressed in the egg cylinder stage just prior to the onset of gastrulation in mouse (Figure 2; [58]). It has been argued by several authors that the common roles and expression patterns of these genes are the result of evolutionary conservation, or evolution by descent [59–61]. We infer from this that the stage of mammalian embryogenesis that is analogous to the frog blastula and the *Drosophila* blastoderm is the pregastrulation stage, at which all of these conserved features of patterning occur. Mouse gastrulation occurs at the egg cylinder stage (6.5 dpc) [37]. Thus, it is the egg cylinder stage that is analogous to the frog blastula and fly blastoderm.

In reviewing the mechanisms involved in embryonic axis specification in Chordates, Eyal-Giladi similarly concluded that the mammalian blastocyst is not the analog of the blastula [62]. She argues that ‘the homology of the germ layers and of the cavities of a mammalian embryo to all other types of amniotic embryos ... is quite clear’, and that it is ‘the narrow slit separating the epiblast from the hypoblast that should be identified as a blastocoele’. This ‘narrow slit’, which develops within the inner cell mass at about the time that the mammalian blastocyst implants, is distinct from the large cavity of the blastocyst, which can be considered an empty yolk vesicle (see below; Figure 2 and 3).

It should be noted that gastrulation in the mouse is often considered to begin with the separation of primitive endoderm from the epiblast around 4.5 dpc. However, the primitive endoderm is homologous to the hypoblast, an extraembryonic tissue. For comparison with other systems we use the more widely accepted definition of gastrulation based on the formation of the embryonic germ layers. In mouse, this occurs in conjunction with primitive streak formation at day 6.5. Given this alignment of embryogenesis, should we look for the mammalian analog of the cleavage stages just before gastrulation or should we look almost a week earlier, immediately after fertilization? We have used gastrulation as our reference.

Peri-Gastrulation Cycles Show Features of Fast Embryonic Cycles

As first noted by Snow, the leisurely pace of cell cycle progression that characterizes early mouse embryogenesis increases dramatically during the egg cylinder stage at 6.5 days [37, 63]. The extraembryonic tissues do not undergo especially rapid cycles, but areas within the

embryonic ectoderm have cell cycle times as short as 2.2 hours. Snow [63] suggested that the divisions are limited to a 'proliferative zone', whereas MacAuley et al. [64] suggested that all cells passing through the primitive streak undergo rapid cycles. Because cells move during gastrulation, the region of high proliferation is not a constant group of cells; rather, many of the embryonic cells pass through a period of rapid division in close coordination with their gastrulation movements. As the primitive streak lengthens across the embryo, a subset of the embryonic ectodermal cells change shape and move out of the ectodermal layer to form the future mesoderm and endoderm. The primitive streak cell population is, therefore, dynamic with ectodermal cells moving in to replace those moving out to form the mesoderm. During this process, cells of the ectoderm proliferate rapidly to populate the primitive streak.

Remarkably, the rapid pre-gastrulation (at the cellular level) or peri-gastrulation (at the embryonic level) divisions in mammalian embryos share many features of pre-gastrulation cell cycles in flies and frogs. First, the feature that identified these cycles, their speed, sets them apart from the cycles of other mammalian cells and is an important parallel to the embryonic cycles in frogs and flies. As mentioned above, cell cycles of the proliferative zone in a 6.5 day old mouse embryo take on average 2.2 hours [37,63]. Rat embryos of a similar stage exhibit cell cycle times of less than 3–3.5 hours [64]. Furthermore, like the cleavage cycles of flies and frogs, the peri-gastrulation cycles of rat embryos show non-existent or short (0–30 min) G1 and G2 phases [64]. Second, cellular growth and division appears uncoupled to a certain extent in the peri-gastrulation cycles of mouse and rat, such that cells of the primitive streak are smaller than those of their predecessors in the ectoderm [64]. This may be an inevitable consequence of a short G1 phase during which cellular growth typically occurs in many somatic cell types. Again, this is a feature that is shared by rapid embryonic divisions of frogs and flies that subdivide the large egg mass. Third, studies in rat show that pre-gastrulation cycles lack a checkpoint that inhibits mitosis in the presence of DNA damage. Rather than execute a cell cycle delay, these cells die [65]. Similarities of these cycles to cleavage cycles of frog and fly with regard to checkpoint control are further discussed below.

New Evidence of Parallels Between the Cleavage Cycles of Frog and Fly and Peri-Gastrulation Cycles of Vertebrates

The repair of DNA damage is tied to progression through the cell cycle. In species from yeast to human, genes have been characterized that arrest progress of the cell cycle to mitosis when DNA is damaged [66]. This coupling gives cells the opportunity to repair the damage prior to irrevocable genetic damage. In all species examined, the genes coupling cell cycle progression to DNA damage are dispensable for undisturbed cell cycle progression – at least in most cycles. Indeed, the genes in this regulatory pathway are the quintessential checkpoint genes – genes whose ability to modulate cell cycle progression is thought to be engaged only when events go awry, as would occur upon irradiation. A highly related checkpoint pathway senses the completion of DNA replication and prevents inappropriate, premature progress to mitosis. Again these genes are generally dispensable in undisturbed cell cycles.

In the context of the prevailing idea that the embryonic cleavage cycles lack checkpoint controls [16,17,19,67], it came as a surprise that analysis of mutations in *Drosophila* ATM/ATR (*mei-41*) and Chk1 (*grapes*) showed that these genes are uniquely required in the early cycles [20,68,69]. Thus, genes that are dispensable when they function in checkpoint control are required in the cycles in which we had inferred they were not active. Clearly they must be active in the cleavage cycles, but what are they doing if the cells do not exhibit a checkpoint? A partial answer comes from more detailed studies of checkpoint action in the early cycles.

In vitro studies in a cycling *Xenopus* extract paralleled findings in the intact embryo in that blocking DNA replication did not block cycling as assessed either by oscillations of cyclin/

Cdk kinase or by entry of nuclei into M phase [19]. However, when the density of nuclei in the extract was increased, the cycling of the extract became dependent on replication [70]. It was inferred that the extract, and presumably the embryo, is capable of coupling mitosis to S-phase but that the signal generated at the very low nuclear to cytoplasmic ratio in the early embryo was not sufficient to inhibit progress to mitosis. Other findings are also consistent with a quantitative interpretation.

Xenopus embryos and the cycling extracts also appear to lack a spindle checkpoint, as the use of drugs blocking spindle formation does not prevent exit from mitosis. However, when the nuclear density was increased in the extract, a drug-induced arrest of mitosis became evident, as was seen in the case of the replication checkpoint. This arrest depends on gene products homologous to those acting in checkpoint regulation of mitotic progression in yeast [67].

In *Drosophila*, it was found that aphidicolin inhibition of S-phase during embryonic cycle 11 or 12 briefly delays the subsequent mitosis [20]. This suggests that the early embryos do have a mechanism that can delay mitosis when replication is incomplete. The transient nature of the block suggests that the mechanism might not have the quantitative ability to fully block the activators of mitosis present during these stages. In addition to these indications of weak checkpoint activity in the models that had originally suggested an absence of checkpoints, there is full-fledged checkpoint activity in some systems. Disruption of the spindle in *Drosophila* embryos arrests cleavage nuclei in metaphase, clam embryos exhibit checkpoint arrests in response to both microtubule depolymerizing drugs and inhibitors of DNA replication, and sea urchin embryos arrest in response to inhibitors of DNA replication [25,71–73]. Thus, it now appears that checkpoint pathways are present during cleavage cycles.

Drosophila mutants in the checkpoint genes *mei-41* (ATM/ATR) and *grapes* (Chk1) are viable, but they are maternal effect lethals, i.e. mutant adult females give no or few progeny [20,68,74]. Embryos from mutant mothers show severely defective cell cycles by the time of mitosis 12. Because the requirement for these functions is seen in the absence of any perturbation, this finding indicates that control of the early cycles is distinct from that of other cell cycles. It is unclear why undisturbed rapid divisions should require checkpoint activities. In cycles without the leeway provided by a G2-phase, these functions are perhaps needed to ensure that mitosis does not initiate until after DNA synthesis is completed [20,74]. Alternatively, the unusually fast mitoses perhaps rely on this pathway to serve a different role that ensures the proper order and timing of mitotic events [69].

Regardless of the mechanism that underlies the unique requirement for these checkpoint genes during the early cleavage stages of *Drosophila*, it is striking that the mouse embryo exhibits a similar requirement at the time of the rapid peri-gastrulation divisions. Analysis of mouse mutations in ATR and Chk1 shows that these genes are dispensable for cell divisions shortly after fertilization, but they become essential in peri-gastrulation mouse embryos. ATR^{-/-} embryos develop to 3.5 dpc, but die between implantation (after 4.5 dpc) and 7.5 dpc, which encompasses the peri-implantation stages under discussion. Chk1^{-/-} embryos die between 3.5 and 7.5 dpc, which again encompasses the peri-gastrulation stages [75,76]. In culture, cells from ATR mutant embryos display defects that are consistent with cell division problems, such as small size, decreased proliferative index and chromosome fragmentation [77]. As in the case of fly embryos, the reason for the requirement for ATR and Chk1 in mouse embryogenesis remains unclear. Nonetheless, it would be interesting to know if ATR and Chk1 homologs play essential roles during rapid cell cycles in other phyla.

The rapid early embryonic cycles of the fly share another unusual feature with the perigastrulation cell cycles of the mouse egg cylinder. During the early cycles in *Drosophila*, embryos are remarkably radiation sensitive. The dose of ionizing radiation that kills 50% of

embryos undergoing cleavage divisions is about 250 Rads, compared to ~4000 Rads in larvae [78,79]. Detailed analysis of the timing of the change in sensitivity showed that embryos develop a much increased radiation tolerance at the close of the rapid mitotic cycles. Real time analysis following irradiation revealed that early embryos fail to arrest progress into mitosis within the cycle in which they are irradiated, and that they show severe defects in subsequent mitosis [68,80]. Interestingly, the cell cycles of gastrulating mouse embryos also lack a checkpoint response that inhibits mitosis in the presence of DNA damage [65]. Rather than execute a cell cycle delay, these cells readily commit cell death, reminiscent of the radiation sensitivity of the cleavage stage fly embryo. Consequently, cells of mouse embryos show higher sensitivity to killing by ionizing radiation during gastrulation than at an earlier stage. Likewise, cells of the frog embryo show higher sensitivity to killing by ionizing radiation during cleavage cycles than at later stages [81]. Although more complete time courses are needed, the results suggest that embryonic cells are unusually sensitive to radiation during the rapid cell cycles. It will also be of interest to determine whether the inability to regulate the entry into mitosis in response to damaged DNA is a feature shared by rapidly cycling embryonic cells of different phyla. While many aspects of the roles of the checkpoint genes in the embryonic cycles need to be defined, it is striking that the early cleavage cycles of model organisms share with the peri-gastrulation cycles of mammals a unique requirement for the checkpoint pathway and an especially high sensitivity to irradiation.

Post-Fertilization Cycles in Mammalian Embryos May Have Been Derived from Pre-Fertilization Processes

If peri-gastrulation cell cycles of mammals are equivalent to the cleavage of early embryos in other metazoa, why are they not seen immediately following fertilization? In mice and humans, divisions immediately after fertilization do not produce the embryo proper but extra-embryonic tissues that will supply nutrients to the embryo. Two generations of primary extraembryonic tissues are produced before the mammalian embryo initiates events of embryogenesis: the post-fertilization divisions produce a blastocyst composed of a sphere of trophoblast (extraembryonic) and inner cell mass (ICM), and later (4 days into mouse embryogenesis) the ICM produces the primitive endoderm (extraembryonic) and the primitive ectoderm (embryonic) [37]. The first generation of extraembryonic tissue, the trophoblast, will differentiate into the placenta and the chorion, while the later formed primitive endoderm/hypoblast will first differentiate into parietal and visceral endoderm and later into the yolk sac (Figure 3). Thus, strictly speaking, the divisions that immediately follow fertilization are not embryonic divisions yet, but rather produce tissues involved in nourishing the embryo.

In most vertebrates, it is the yolk sac that performs the nutritive role. In meroblastic embryos, which are derived from yolk laden eggs and have incomplete early cytokinesis (e.g. chick), the yolk sac, true to its name, sits at the interface with the abundant yolk and harvests yolk material to provide for the embryo. These organisms lack the first wave of extraembryonic tissue generation and show no evidence of a trophoblast [62,82]. The primitive placenta of sharks and viviparous reptiles is formed by a secondary specialization of the yolk sac which serves a dual role of providing nutrients first from the yolk and then from the mother. While the yolk sac retains some nutritive functions in mammals in which it functions to take up material from uterine fluid, the trophoblast-derived placenta performs the major nutritive role. It appears as if evolution has added a new stage to early embryogenesis in order to generate this nutritive organ. It is of interest to consider whether the trophoblast had an evolutionary precursor and if so, what it might be.

Given arguments that the cavity of the blastocyst is analogous to an empty yolk vesicle (see above; Figure 2,3), perhaps the evolutionary precursors of the trophoblast cells will surround the yolk mass in the predecessors of mammals. In mammals, the cavity of the blastocyst comes

to be surrounded by two layers of cells; the shell of trophoblastic cells defines this cavity and cells derived from the primitive endoderm/hypoblast migrate over the inner surface of the trophoblast to form the second layer, the yolk sac. In the embryos of birds, hypoblast cells migrate over the surface of the yolk to form the yolk sac, but there is no overlying tissue analogous to the trophoblast. However, earlier, during oogenesis, there are maternally derived granulosa cells surrounding the yolk. These cells comprise a monolayered epithelium that contributes to the extraordinary accumulation of yolk during oogenesis. In summary, it appears that the trophoblast is a novel feature of embryogenesis that was added during the evolutionary history of mammals. Birds and presumably reptiles, though lacking any obvious zygotic analog of the trophoblast, possess maternal tissue that is spatially and physically analogous.

How was the developmental program modified to introduce a new stage and accommodate the trophoblast? The above comparison to chick as well as broader evolutionary considerations suggest how this change may have occurred. In contrast to mammalian eggs, the eggs of non-uterine animals grow to very large sizes during oogenesis. The huge expansion of the oocyte is promoted by specialized nutritive tissues. Detailed analysis, largely outside chordates, shows that there are two categories of nutritive cells, nurse cells and follicle cells. In diverse organisms, nurse cells are sister cells of oocytes (e.g., the beetle, *Dytiscus* or the leech, *Pisciola*; [21]). In chordates, the differentiating oocytes appear to recruit cells that become the granulosa cells or follicle cells that perform this nutritive role [83]. However, the lineages that give rise to granulosa cells in different chordates and their relationship to the nurse cells and follicle cells of non-chordate species are ill defined. We speculate that in non-mammalian chordates at least some of the granulosa cells are sister cells of the oocyte, much like the nurse cells in other phyla. If, during the evolution of mammals, a change in the timing of events occurred, so that meiosis and fertilization shifted to precede the cell divisions that separated the later derived granulosa cells from the oocyte, some of the granulosa cells, namely the one's produced late in the lineage, would then be produced after fertilization. According to this scenario, these zygotically produced 'granulosa cells' would be the trophoblast cells. They would continue to perform a nutritive function but now would be providing nutrition to the developing embryo rather than the oocyte. Though speculative, this hypothesis explains features of early mammalian development beyond the origins of the trophoblast. For example, the transition to the small size of the mammalian egg could be explained by its 'precocious' maturation. Furthermore, the switch from maternally regulated to zygotically regulated early development would follow from the change in relative time of fertilization, which would switch many events from pre-fertilization to post-fertilization. An adaptation leading to viviparous fish reveals another case in which the relative timing of fertilization is altered. In some viviparous fish (some Poeciliid teleosts), the egg is fertilized within the follicle and is supported through embryonic development within the follicle prior to 'ovulation'/birth [84].

Our proposal would predict that early zygotic development of mammals might resemble steps in follicular development in monotremes (e.g. *Platypus*), birds, reptiles and other non-therian species. In the formation of chordate follicles, the developing oocyte and a few 'granulosa cells' are isolated within a basement membrane, the granulosa cells begin to form an epithelium on the outside and the oocyte grows on the inside (e.g. [85]). One can make analogies between this stage and the morula of mammalian embryogenesis, a stage at which the central cells commit to an embryonic fate and the outer cells begin to take on epithelial features of the trophoblast. Other similarities can be drawn between the 'granulosa cell' layer and the development of the trophoblast; however, if they are related by descent, modification of subsequent events gives these tissues distinct roles.

Despite the enormous evolutionary distance, features of the developmental programs in the nurse cells of non-chordate species might be taken as support for our proposal. As mentioned, the nurse cells are produced from germ cell precursors and hence are sister cells of the oocytes

[21]. In some cases the fates of nurse cells and the oocytes are separated extremely late. Indeed, in the leech *Piscicola* the nurse cells enter the meiotic divisions. If the sibling relationship of nurse and germ cells were to persist only slightly longer, the lineages might separate after fertilization, as we suggest for mammals. Furthermore, throughout most of metazoan phylogeny, nurse cells contribute to the growth of the egg and accumulation of maternally provided yolk stores. The granulosa cells of non-mammalian chordates make a similar contribution, while trophoblast cells transfer nutrients to the embryo, a role not unlike that of nurse/granulosa cells. Additionally, nurse cells in diverse species grow to very large sizes and develop huge polyploid nuclei. The enormous, branched nucleus of the nurse cells of an earwig provides a dramatic example of this [21]. The nurse cells of *Drosophila* and other species are polyploid and produce prodigious levels of RNA (e.g., the polychaete *Ophryotrocha labronica* and the dipteran, *Calliphora* [86,87]). As suggested by their name, the trophoblastic giant cells are very large, and they also endocycle to become polyploid. Perhaps this unusual behavior of the trophoblast cells has a primordial origin in nurse cell developmental programs that evolved in non-chordate predecessors.

According to the proposal made here, the early cell cycles of mammalian eggs would not be analogous to the cleavage cycles in other embryos, but to cell divisions producing the oocyte and its sister cells. While we have been unable to locate, among published works, descriptions of these events in the non-mammalian chordates, these events are well known in *Drosophila*. Four cell divisions of a precursor cell produce the oocyte and the 15 nurse cells that populate a *Drosophila* egg chamber. These four egg chamber divisions take on average 6 hours per cell cycle, much longer than early cleavage cycles [88]. Based on our proposal that the timing of fertilization is displaced with respect to other events in mammals, we propose the speculative alignment shown in Figure 2.

A Precedent for Our Proposal

Biology provides an independent example of deferred development and of production of extraembryonic tissues. Like the embryos of uterine animals, the embryos of parasitic wasps develop in a highly nutritive environment. After fertilization, the eggs of *Copidosoma floridanum* undergo dramatic growth and proliferation to produce multiple twin embryos within the hemolymph of parasitised caterpillars [89]. To accomplish this, embryonic patterning and gastrulation are deferred, in a similar manner as in mammalian development. Additionally, these embryos possess extraembryonic membranes [90]. These membranes develop from polar nuclei, the reduction nuclei of meiosis that are usually discarded. In this organism, it is unambiguous that the events represent a deferral of embryonic development because the later patterning of the multiple twinned embryos can be clearly aligned with the developmental programs of other insects. Thus, it is undeniable that deferral of early embryonic programs evolved at least once. We propose that mammals represent a second example.

The establishment of analogies between model systems and mammalian development extends the impact of studies in model organisms. In the view presented in this article, the studies of rapid cell cycles in frogs and flies will directly benefit the understanding of mammalian embryonic cell cycles.

Acknowledgments

We thank Gail Martin, Gilles R. Hickson, Pascale Dickers, Joan Ruderman, and Michael Stowell for helpful comments.

References

1. O'Farrell, PH. How metazoans reach their final size: the natural history of bigness. In: Hall, M.; Raff, M.; Thomas, G., editors. Cell Growth. Cold Spring Harbor Laboratory Press; 2003.

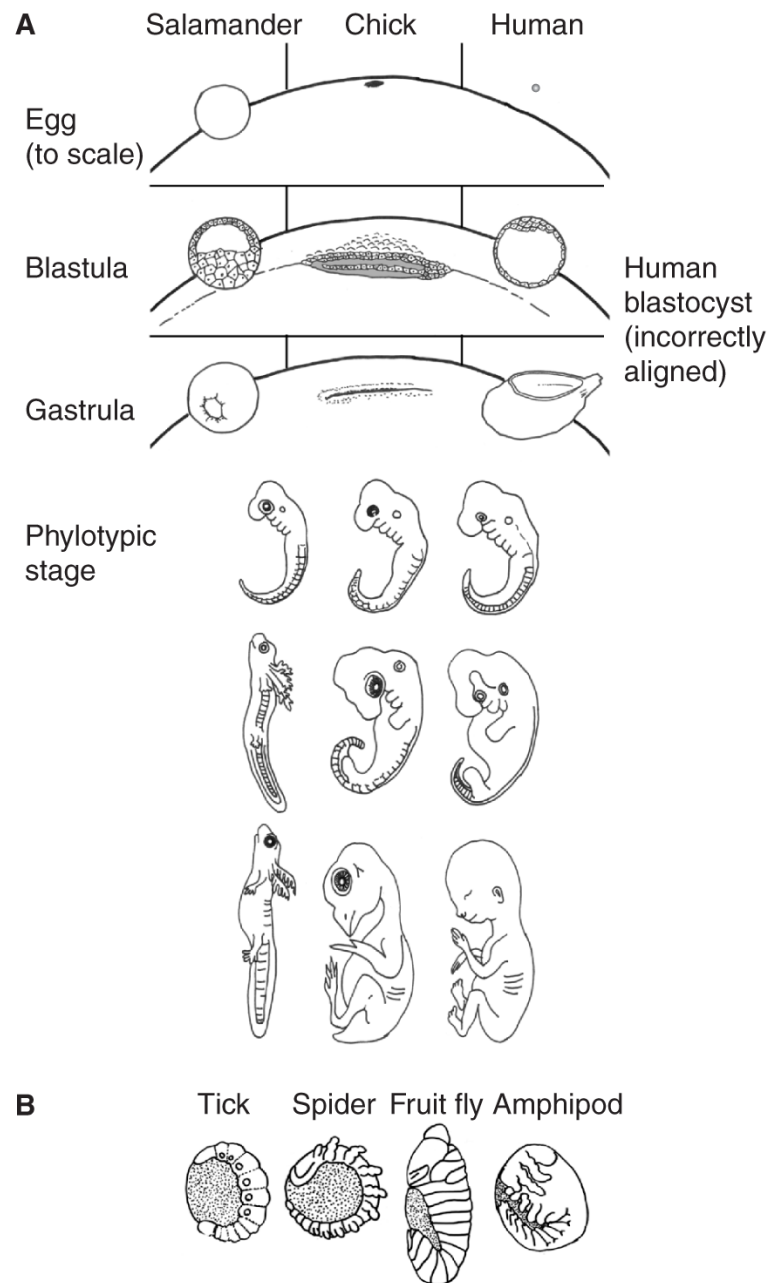
2. Alberts, B.; Johnson, A.; Lewis, J.; Raff, M.; Roberts, K.; Walter, P. *Molecular Biology of the Cell*. Vol. 4. Garland Science; 2002.
3. Foe, VE.; Odell, GM.; Edgar, BA. Mitosis and morphogenesis in the *Drosophila* embryo. In: Bate, M.; Martinez Arias, A., editors. *The Development of Drosophila melanogaster*. Cold Spring Harbor Laboratory Press; 1993.
4. Milan M, Campuzano S, Garcia-Bellido A. Cell cycling and patterned cell proliferation in the *Drosophila* wing during metamorphosis. *Proc Natl Acad Sci USA* 1996;93:11687–11692. [PubMed: 8876197]
5. Milan M, Campuzano S, Garcia-Bellido A. Cell cycling and patterned cell proliferation in the wing primordium of *Drosophila*. *Proc Natl Acad Sci USA* 1996;93:640–645. [PubMed: 8570608]
6. Mitchison, JM. *The Biology of the Cell Cycle*. Cambridge University Press; 1971.
7. Conlon I, Raff M. Differences in the way a mammalian cell and yeast cells coordinate cell growth and cell-cycle progression. *J Biol* 2003;2:7. [PubMed: 12733998]
8. Ferrell JE Jr, Wu M, Gerhart JC, Martin GS. Cell cycle tyrosine phosphorylation of p34cdc2 and a microtubule-associated protein kinase homolog in *Xenopus* oocytes and eggs. *Mol Cell Biol* 1991;11:1965–1971. [PubMed: 2005892]
9. Newport J, Kirschner M. A major developmental transition in early *Xenopus* embryos: I. characterization and timing of cellular changes at the midblastula stage. *Cell* 1982;30:675–686. [PubMed: 6183003]
10. Newport JW, Kirschner MW. Regulation of the cell cycle during early *Xenopus* development. *Cell* 1984;37:731–742. [PubMed: 6378387]
11. Baldari CT, Amaldi F. DNA reassociation kinetics in relation to genome size in four amphibian species. *Chromosoma* 1976;59:13–22. [PubMed: 826380]
12. Blumenthal AB, Kriegstein HJ, Hogness DS. The units of DNA replication in *Drosophila melanogaster* chromosomes. *Cold Spring Harb Symp Quant Biol* 1974;38:205–223. [PubMed: 4208784]
13. McKnight SL, Miller OL Jr. Electron microscopic analysis of chromatin replication in the cellular blastoderm *Drosophila melanogaster* embryo. *Cell* 1977;12:795–804. [PubMed: 411576]
14. Spradling A, Orr-Weaver T. Regulation of DNA replication during *Drosophila* development. *Annu Rev Genet* 1987;21:373–403. [PubMed: 3327470]
15. Walter J, Newport JW. Regulation of replicon size in *Xenopus* egg extracts. *Science* 1997;275:993–995. [PubMed: 9020085]
16. Raff JW, Glover DM. Nuclear and cytoplasmic mitotic cycles continue in *Drosophila* embryos in which DNA synthesis is inhibited with aphidicolin. *J Cell Biol* 1988;107:2009–2019. [PubMed: 3143733]
17. Raff JW, Glover DM. Centrosomes, and not nuclei, initiate pole cell formation in *Drosophila* embryos. *Cell* 1989;57:611–619. [PubMed: 2497990]
18. Glover DM, Alphey L, Axton JM, Cheshire A, Dalby B, Freeman M, Girdham C, Gonzalez C, Karess RE, Leibowitz MH, et al. Mitosis in *Drosophila* development. *J Cell Sci Suppl* 1989;12:277–291. [PubMed: 2534556]
19. Newport J, Dasso M. On the coupling between DNA replication and mitosis. *J Cell Sci Suppl* 1989;12:149–160. [PubMed: 2517560]
20. Sibon OC, Stevenson VA, Theurkauf WE. DNA-replication checkpoint control at the *Drosophila* midblastula transition. *Nature* 1997;388:93–97. [PubMed: 9214509]
21. Wilson, EB. *The cell in development and heredity*. Vol. 3. New York: The MacMillan Company; 1925.
22. Bissen ST, Weisblat DA. The durations and compositions of cell cycles in embryo of the leech *Helobdella triserialis*. *Development* 1989;106:105–118.
23. Parisi E, Filosa S, De Petrocellis B, Monroy A. The pattern of cell division in the early development of the sea urchin *Paracentrotus lividus*. *Dev Biol* 1978;65:38–49. [PubMed: 680360]
24. Wood, WB. Embryology. In: Wood, WB., editor. *The Nematode Caenorhabditis elegans*. Cold Spring Harbor Laboratory Press; 1988.

25. Hunt T, Luca FC, Ruderman JV. The requirements for protein synthesis and degradation, and the control of destruction of cyclins A and B in the meiotic and mitotic cell cycles of the clam embryo. *J Cell Biol* 1992;116:707–724. [PubMed: 1530948]
26. Wright SJ, Schatten G. Teniposide, a topoisomerase II inhibitor, prevents chromosome condensation and separation but not decondensation in fertilized surf clam (*Spisula solidissima*) oocytes. *Dev Biol* 1990;142:224–232. [PubMed: 2172057]
27. Tudge, C. *The Variety of Life*. Oxford University Press; 2000.
28. Eyal-Giladi H, Kochav S. From cleavage to primitive streak formation: a complementary normal table and a new look at the first stages of the development of the chick. I General morphology. *Dev Biol* 1976;49:321–337. [PubMed: 944662]
29. Grosshans J, Wieschaus E. A genetic link between morphogenesis and cell division during formation of the ventral furrow in *Drosophila*. *Cell* 2000;101:523–531. [PubMed: 10850494]
30. Mata J, Curado S, Ephrussi A, Rorth P. Tribbles coordinates mitosis and morphogenesis in *Drosophila* by regulating string/CDC25 proteolysis. *Cell* 2000;101:511–522. [PubMed: 10850493]
31. Seher TC, Leptin M. Tribbles, a cell-cycle brake that coordinates proliferation and morphogenesis during *Drosophila* gastrulation. *Curr Biol* 2000;10:623–629. [PubMed: 10837248]
32. Shermoen AW, O'Farrell PH. Progression of the cell cycle through mitosis leads to abortion of nascent transcripts. *Cell* 1991;67:303–310. [PubMed: 1680567]
33. Edgar BA, Schubiger G. Parameters controlling transcriptional activation during early *Drosophila* development. *Cell* 1986;44:871–877. [PubMed: 2420468]
34. Hartenstein V, Campos-Ortega J. Fate-mapping in wild-type *Drosophila melanogaster*. I The spatio-temporal pattern of embryonic cell divisions. *Wilhelm Roux Arch Dev Biol* 1985;194:181–195.
35. Hartenstein V, Rudloff E, Campos-Ortega J. The pattern of proliferation of the neuroblasts in the wild-type embryo of *Drosophila melanogaster*. *Wilhelm Roux Arch Dev Biol* 1987;196:473–485.
36. Patel NH, Ball EE, Goodman CS. Changing role of even-skipped during the evolution of insect pattern formation. *Nature* 1992;357:339–342. [PubMed: 1350328]
37. Hogan, B.; Beddington, R.; Costantini, F.; Lacy, E. *Manipulating the Mouse Embryo*. Vol. 2. Cold Spring Harbor Laboratory Press; 1994.
38. Fulka J Jr, First NL, Fulka J, Moor RM. Checkpoint control of the G2/M phase transition during the first mitotic cycle in mammalian eggs. *Hum Reprod* 1999;14:1582–1587. [PubMed: 10357980]
39. Thomas JB, Bastiani MJ, Bate M, Goodman CS. From grasshopper to *Drosophila*: a common plan for neuronal development. *Nature* 1984;310:203–207. [PubMed: 6462206]
40. Patel NH, Kornberg TB, Goodman CS. Expression of engrailed during segmentation in grasshopper and crayfish. *Development* 1989;107:201–212. [PubMed: 2632219]
41. Gould, SJ. *Ontogeny and Phylogeny*. Belknap Press; 1977.
42. Holley SA, Neul JL, Attisano L, Wrana JL, Sasai Y, O'Connor MB, De Robertis EM, Ferguson EL. The *Xenopus* dorsalizing factor noggin ventralizes *Drosophila* embryos by preventing DPP from activating its receptor. *Cell* 1996;86:607–617. [PubMed: 8752215]
43. Bachiller D, Klingensmith J, Shneyder N, Tran U, Anderson R, Rossant J, De Robertis EM. The role of chordin/Bmp signals in mammalian pharyngeal development and DiGeorge syndrome. *Development* 2003;130:3567–3578. [PubMed: 12810603]
44. Garcia Abreu J, Coffinier C, Larrain J, Oelgeschlager M, De Robertis EM. Chordin-like CR domains and the regulation of evolutionarily conserved extracellular signaling systems. *Gene* 2002;287:39–47. [PubMed: 11992721]
45. Oelgeschlager M, Reversade B, Larrain J, Little S, Mullins MC, De Robertis EM. The pro-BMP activity of Twisted gastrulation is independent of BMP binding. *Development* 2003;130:4047–4056. [PubMed: 12874126]
46. Re'em-Kalma Y, Lamb T, Frank D. Competition between noggin and bone morphogenetic protein 4 activities may regulate dorsalization during *Xenopus* development. *Proc Natl Acad Sci USA* 1995;92:12141–12145. [PubMed: 8618860]
47. Isaacs HV, Andreazzoli M, Slack JM. Anteroposterior patterning by mutual repression of orthodenticle and caudal-type transcription factors. *Evol Dev* 1999;1:143–152. [PubMed: 11324099]

48. Stott D, Kispert A, Herrmann BG. Rescue of the tail defect of Brachyury mice. *Genes Dev* 1993;7:197–203. [PubMed: 8436292]
49. Kispert A, Herrmann BG, Leptin M, Reuter R. Homologs of the mouse Brachyury gene are involved in the specification of posterior terminal structures in *Drosophila*, *Tribolium*, and *Locusta*. *Genes Dev* 1994;8:2137–2150. [PubMed: 7958884]
50. Woollard A, Hodgkin J. The *Caenorhabditis elegans* fate-determining gene mab-9 encodes a T-box protein required to pattern the posterior hindgut. *Genes Dev* 2000;14:596–603. [PubMed: 10716947]
51. Schulte-Merker S, van Eeden FJ, Halpern ME, Kimmel CB, Nusslein-Volhard C. no tail (ntl) is the zebrafish homologue of the mouse T (Brachyury) gene. *Development* 1994;120:1009–1015. [PubMed: 7600949]
52. Giovannini N, Rungger D. Antisense inhibition of Xbrachyury impairs mesoderm formation in *Xenopus* embryos. *Dev Growth Differ* 2002;44:147–159. [PubMed: 11940101]
53. De Roberts EM, Blum M, Niehrs C, Steinbeisser H. Goosecoid and the organizer. *Dev Suppl* 1992;1:167–171. [PubMed: 1363720]
54. Blum M, Gaunt SJ, Cho KW, Steinbeisser H, Blumberg B, Bittner D, De Robertis EM. Gastrulation in the mouse: the role of the homeobox gene goosecoid. *Cell* 1992;69:1097–1106. [PubMed: 1352187]
55. Kispert A, Ortnier H, Cooke J, Herrmann BG. The chick Brachyury gene: developmental expression pattern and response to axial induction by localized activin. *Dev Biol* 1995;168:406–415. [PubMed: 7729577]
56. Shoguchi E, Satoh N, Maruyama YK. A starfish homolog of mouse T-brain-1 is expressed in the archenteron of *Asterina pectinifera* embryos: possible involvement of two T-box genes in starfish gastrulation. *Dev Growth Differ* 2000;42:61–68. [PubMed: 10831044]
57. Gross JM, McClay DR. The role of Brachyury (T) during gastrulation movements in the sea urchin *Lytechinus variegatus*. *Dev Biol* 2001;239:132–147. [PubMed: 11784024]
58. Wilkinson DG, Bhatt S, Herrmann BG. Expression pattern of the mouse T gene and its role in mesoderm formation. *Nature* 1990;343:657–659. [PubMed: 1689462]
59. McGhee JD. Homologous tails? Or tales of homology? *Bioessays* 2000;22:781–785. [PubMed: 10944579]
60. Adell T, Grebenjuk VA, Wiens M, Muller WE. Isolation and characterization of two T-box genes from sponges, the phylogenetically oldest metazoan taxon. *Dev Genes Evol*. 2003in press
61. De Robertis EM, Fainsod A, Gont LK, Steinbeisser H. The evolution of vertebrate gastrulation. *Dev Suppl* 1994;1:117–124. [PubMed: 7579512]
62. Eyal-Giladi H. Establishment of the axis in chordates: facts and speculations. *Development* 1997;124:2285–2296. [PubMed: 9199355]
63. Snow MHL. Gastrulation in the mouse: growth and regionalization of the epiblast. *J Exp Emryol Exp Morphol* 1977;42:293–303.
64. MacAuley A, Werb Z, Mirkes PE. Characterization of the unusually rapid cell cycles during rat gastrulation. *Development* 1993;117:873–883. [PubMed: 8325243]
65. Heyer BS, MacAuley A, Behrendtsen O, Werb Z. Hypersensitivity to DNA damage leads to increased apoptosis during early mouse development. *Genes Dev* 2000;14:2072–2084. [PubMed: 10950870]
66. Elledge SJ. Cell cycle checkpoints: preventing an identity crisis. *Science* 1996;274:1664–1672. [PubMed: 8939848]
67. Minshull J, Sun H, Tonks NK, Murray AW. A MAP kinase-dependent spindle assembly checkpoint in *Xenopus* egg extracts. *Cell* 1994;79:475–486. [PubMed: 7954813]
68. Fogarty P, Campbell SD, Abu-Shumays R, Phalle BS, Yu KR, Uy GL, Goldberg ML, Sullivan W. The *Drosophila* grapes gene is related to checkpoint gene chk1/rad27 and is required for late syncytial division fidelity. *Curr Biol* 1997;7:418–426. [PubMed: 9197245]
69. Su TT, Campbell SD, O'Farrell PH. *Drosophila* grapes/CHK1 mutants are defective in cyclin proteolysis and coordination of mitotic events. *Curr Biol* 1999;9:919–922. [PubMed: 10469601]
70. Dasso M, Newport JW. Completion of DNA replication is monitored by a feedback system that controls the initiation of mitosis in vitro: studies in *Xenopus*. *Cell* 1990;61:811–823. [PubMed: 2160859]

71. Edgar BA, Sprenger F, Duronio RJ, Leopold P, O'Farrell PH. Distinct molecular mechanism regulate cell cycle timing at successive stages of *Drosophila* embryogenesis. *Genes Dev* 1994;8:440–452. [PubMed: 7510257]
72. Ikegami S, Taguchi T, Ohashi M, Oguro M, Nagano H, Mano Y. Aphidicolin prevents mitotic cell division by interfering with the activity of DNA polymerase- α . *Nature* 1978;275:458–460. [PubMed: 692726]
73. Su TT, Sprenger F, DiGregorio PJ, Campbell SD, O'Farrell PH. Exit from mitosis in *Drosophila* syncytial embryos requires proteolysis and cyclin degradation, and is associated with localized dephosphorylation. *Genes Dev* 1998;12:1495–1503. [PubMed: 9585509]
74. Sibon OC, Laurencon A, Hawley R, Theurkauf WE. The *Drosophila* ATM homologue Mei-41 has an essential checkpoint function at the midblastula transition. *Curr Biol* 1999;9:302–312. [PubMed: 10209095]
75. Liu Q, Guntuku S, Cui XS, Matsuoka S, Cortez D, Tamai K, Luo G, Carattini-Rivera S, DeMayo F, Bradley A, et al. Chk1 is an essential kinase that is regulated by Atr and required for the G(2)/M DNA damage checkpoint. *Genes Dev* 2000;14:1448–1459. [PubMed: 10859164]
76. Takai H, Tominaga K, Motoyama N, Minamishima YA, Nagahama H, Tsukiyama T, Ikeda K, Nakayama K, Nakanishi M. Aberrant cell cycle checkpoint function and early embryonic death in Chk1(–/–) mice. *Genes Dev* 2000;14:1439–1447. [PubMed: 10859163]
77. Brown EJ, Baltimore D. ATR disruption leads to chromosomal fragmentation and early embryonic lethality. *Genes Dev* 2000;14:397–402. [PubMed: 10691732]
78. Wurgler, FE.; Ulruch, H. Radiosensitivity of embryonic stages. In: Ashburner, M.; Novitski, E., editors. *The Genetics and Biology of Drosophila*. Academic Press; 1976.
79. Ashburner, M. *Drosophila: A Laboratory Handbook*. Cold Spring Harbor Laboratory Press; 1989.
80. Sibon OC, Kelkar A, Lemstra W, Theurkauf WE. DNA-replication/DNA-damage-dependent centrosome inactivation in *Drosophila* embryos. *Nat Cell Biol* 2000;2:90–95. [PubMed: 10655588]
81. Anderson JA, Lewellyn AL, Maller JL. Ionizing radiation induces apoptosis and elevates cyclin A1-Cdk2 activity before but not after the midblastula transition in *Xenopus*. *Mol Biol Cell* 1997;8:1195–1206. [PubMed: 9243501]
82. Solnica-Krezel L. Pattern formation in zebrafish—fruitful liaisons between embryology and genetics. *Curr Top Dev Biol* 1999;41:1–35. [PubMed: 9784971]
83. Rothchild I. The yolkless egg and the evolution of eutherian viviparity. *Biol Reprod* 2003;68:337–357. [PubMed: 12533395]
84. Thibault R, Shultz R. Reproductive adaptations among viviparous fishes (Cyprinodontiformes and Poecelidae). *Evolution* 1978;32:320–333.
85. Prisco M, Loredana R, Piero A. Ultrastructural studies on developing follicles of the spotted ray *Torpedo marmorata*. *Mol Reprod Dev* 2002;61:78–86. [PubMed: 11774378]
86. Emanuelsson H. Autoradiographic analysis of RNA synthesis in the oocyte-nurse cell complex of the polychaete *Ophryotrocha labronica*. *J Embryol Exp Morphol* 1985;88:249–263. [PubMed: 2416863]
87. Ribbert D. Chromomeres and puffing in experimentally induced polytene chromosomes of *Calliphora erythrocephala*. *Chromosoma* 1979;74:269–298. [PubMed: 510083]
88. Spradling, A. Dev. Genet. of Oogenesis. In: Bate, M.; Martinez Arias, A., editors. *The Development of Drosophila melanogaster*. Vol. 1. Cold Spring Harbor Laboratory Press; 1993.
89. Grbic M, Nagy LM, Carroll SB, Strand M. Polyembryonic development: insect pattern formation in a cellularized environment. *Development* 1996;122:795–804. [PubMed: 8631257]
90. Corley LS, Strand MR. Evasion of encapsulation by the polyembryonic parasitoid *Copidosoma floridanum* is mediated by a polar body-derived extraembryonic membrane. *J Invertebr Pathol* 2003;83:86–89. [PubMed: 12725817]
91. Kuratani S, Nobusada Y, Horigome N, Shigetani Y. Embryology of the lamprey and evolution of the vertebrate jaw: insights from molecular and developmental perspectives. *Philos Trans R Soc Lond B Biol Sci* 2001;356:1615–1632. [PubMed: 11604127]
92. Slack JM, Holland PW, Graham CF. The zootype and the phylotypic stage. *Nature* 1993;361:490–492. [PubMed: 8094230]

93. Patel NH, Martin-Blanco E, Coleman KG, Poole SJ, Ellis MC, Kornberg TB, Goodman CS. Expression of engrailed proteins in arthropods, annelids, and chordates. *Cell* 1989;58:955–968. [PubMed: 2570637]
94. Richardson MK. Vertebrate evolution: the developmental origins of adult variation. *Bioessays* 1999;21:604–613. [PubMed: 10472187]
95. Gilbert, S. *Developmental Biololgy*. Vol. 3. Sunderland: 1991.
96. Elinson, R. Changes in developmental patterns: embryos of amphibians with large eggs. In: Raff, R.; Raff, E., editors. *Development as an evolutionary process*. Alan R. Liss; 1987.
97. Abele, LG., editor. *Embryology, Morphology and Genetics*. Academic Press; 1982.
98. Foelix, RF. *Biology of Spiders*. Oxford University Press; 1996.
99. Sonenshine, DE. *Biology of Ticks*. Oxford University Press; 1991.
100. Behringer RR, Wakamiya M, Tsang TE, Tam PP. A flattened mouse embryo: leveling the playing field. *Genesis* 2000;28:23–30. [PubMed: 11020713]

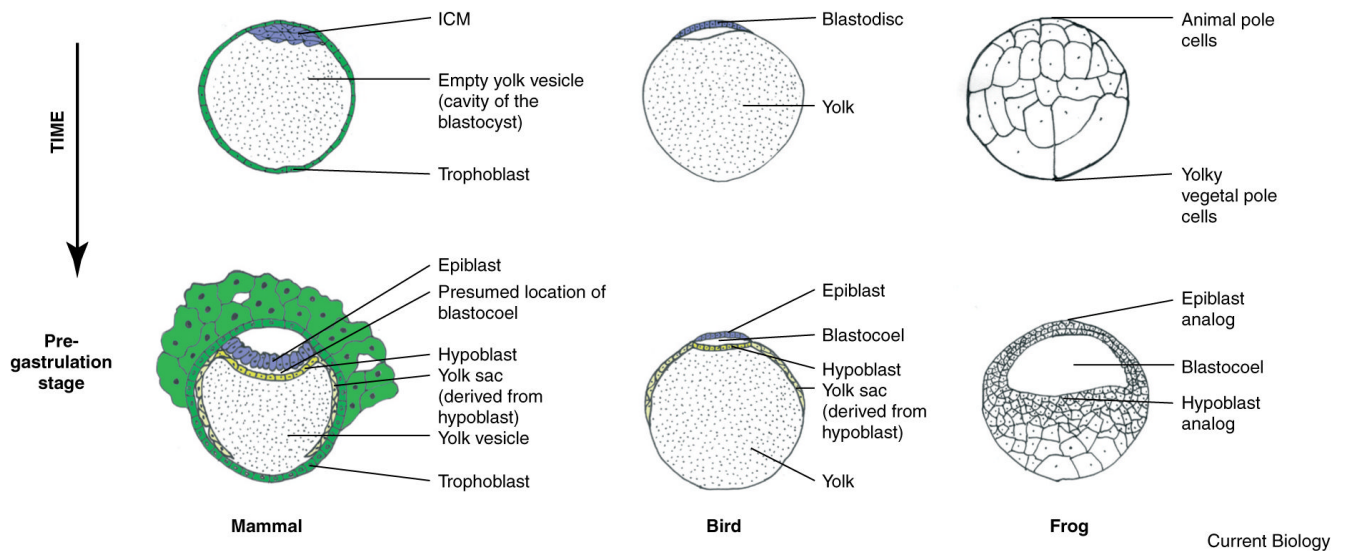


Current Biology

Figure 1. Phylotypic stages of chordates and arthropods

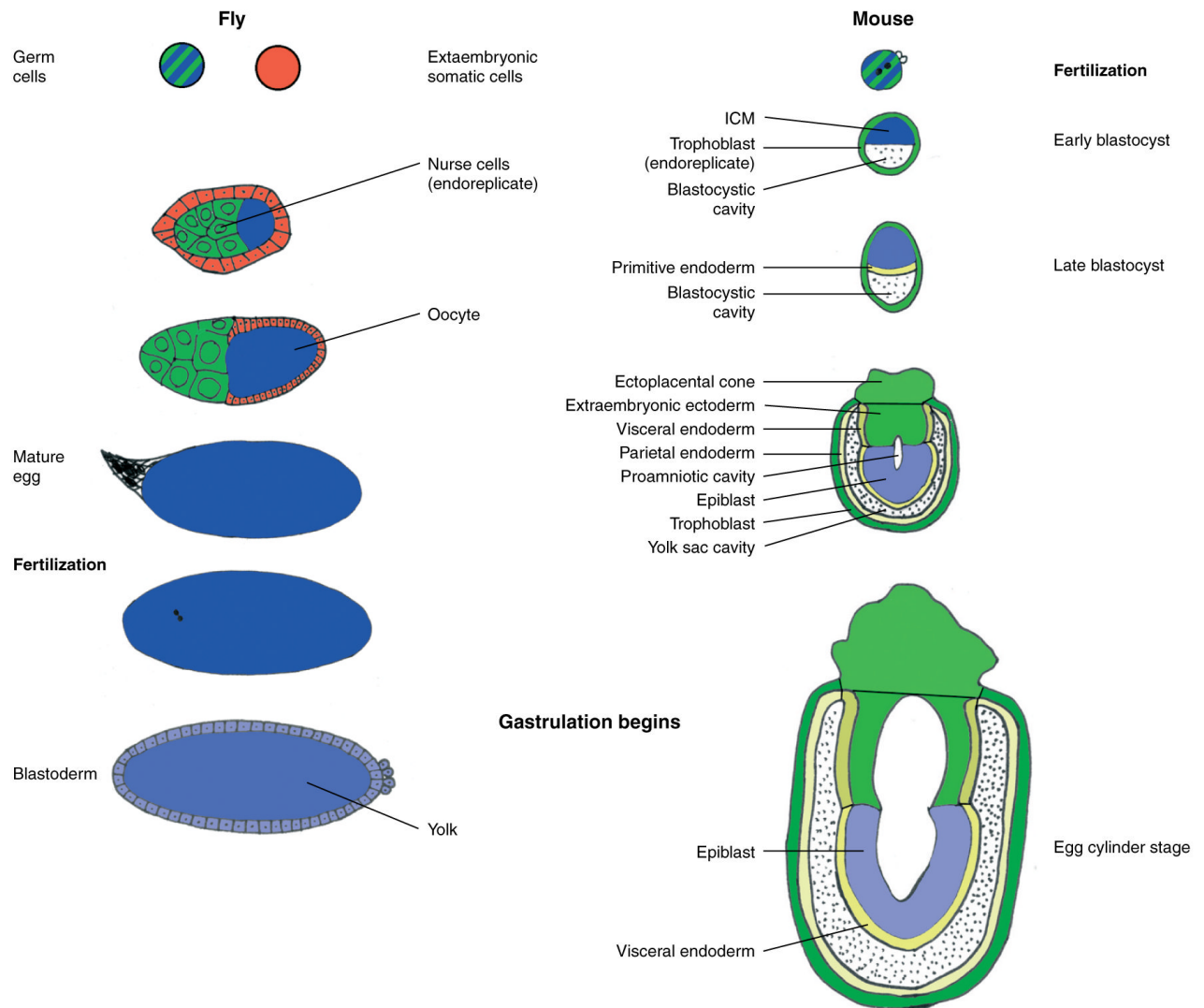
(A) The morphology of chordate embryos, represented by salamander, chick, and human embryos, initially converges to look more similar at the so called pharyngula stage or phylotypic stage [91,92] and subsequently diverges. The high degree of similarity between organisms allows an unambiguous alignment of stages near the phylotypic stage, but ambiguities in alignment can occur at other stages. We argue that the illustrated and generally accepted alignment of the human blastocyst with the blastula stages of amphibian and chick is incorrect (see Figure 2 for a more correct alignment). (B) The phylotypic stage of arthropods, represented by tick, spider, fruit fly and amphipod (fresh water shrimp) embryos, is proposed to occur just after gastrulation [39,93]. Parallels in body segmentation and the position of the

yolk are apparent. Anterior is up. The existence of the phylotypic stage has been questioned by Richardson who pointed out exaggerated homologies in Haeckel's drawings [94]. We do not, however, feel that this is significant criticism of the concept of the phylotypic stage, during which the similarities between embryos across a phyla remain apparent relative to stages before and after. Figure adapted from [95–99] and Flybase at <http://flybase.bio.indiana.edu/>).



Current Biology

Figure 2. Alignment of early embryonic stages among three chordates: frog, chick and human Pregastrulation stages (bottom row) of mammal, bird and amphibian as represented by human, chick and *Xenopus*, show parallels. Cells of the human epiblast (blue) will begin gastrulation into the space between the epiblast and the hypoblast (yellow); cells of the chick epiblast (blue) will likewise move into the space between the epiblast and the hypoblast (yellow). In frogs, cells of the animal cap, which are proposed to be analogous to the epiblast [62], will gastrulate into a space between the animal cap and a subpopulation of vegetal cells, which are proposed to be analogous to the hypoblast cells [62]. The cavity into which human epiblast will gastrulate is, therefore, equivalent to the blastocoelic cavity of chick and frog, but is not named as such. Rather, the yolk vesicle that lies beneath the hypoblast is referred to, incorrectly we propose, as the cavity of the blastocoel. This (incorrect) nomenclature correlates with the (incorrect) alignment of the human blastocyst (top row) with chick and frog blastulae (bottom row). We draw an additional parallel in the migration of hypoblast cells (pale yellow) to form the yolk sac that surrounds the yolk vesicle (in humans) and the yolk (in chick). In an analogous process, cells of the primitive endoderm (yellow), which are equivalent to the hypoblast in mouse (Figure 3) differentiate into visceral endoderm that forms the yolk sac. Figure adapted from [62,95].



Current Biology

Figure 3. Aligning fly and mouse development at gastrulation rather than at fertilization

During *Drosophila* oogenesis (left), the germ cell lineage (green and blue stripes) produces both the extraembryonic nurse cells (green) and the oocyte (blue). The somatic lineage (red) produces follicle cells that surround the developing oocyte. Fertilization in *Drosophila* is followed by 13 rapid nuclear divisions within a common cytoplasm. Cellularization follows to produce a cellular blastoderm in which cortical cells enclose a central yolk mass. Gastrulation ensues next.

In the mouse (right), the fertilized egg (green and blue striped) splits into extra-embryonic lineages (green) and the inner cell mass (ICM; dark blue), which will subsequently shed additional rounds of extra-embryonic lineages (yellow) as well as produce the embryonic epiblast (light blue). Early specified extra-embryonic lineages (green) include the trophoblast cells, which endoreplicate like the nurse cells of *Drosophila*. Extra-embryonic primitive endoderm (yellow in late blastocyst stage), which delaminates from ICM is equivalent to the hypoblast of chick and human (yellow in Figure 2). Primitive endoderm will further differentiate into parietal endoderm and visceral endoderm. At the egg cylinder stage, epiblast cells will migrate during gastrulation into the space between the epiblast and visceral endoderm

(hypoblast equivalent). This is therefore equivalent to the space between epiblast and hypoblast into which epiblast cells move during gastrulation in chicken and humans (Figure 2). Embryonic ectodermal cells of the epiblast exhibit rapid division cycles prior to gastrulation, just as nuclei of *Drosophila* syncytium exhibit rapid divisions prior to cellularization and gastrulation. We do not wish to propose a one-to-one alignment of the fly developmental stages (left) to mouse developmental stages (right). Rather, we propose an alignment at gastrulation in accord with other authors, and suggest the absence of correspondence at fertilization. While we depict an approximate alignment of early stages, we expect that divergence at stages distant from the phylotypic stage has altered the coordination of different events to the point that precise alignment will change depending on the criterion used to assess developmental stage. Nonetheless, this alignment differs significantly from prior alignments at the stage of fertilization in flies and mice. (Figures adapted from [88,95,100].)